

Reconciliation Database: Choice and Cost

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Recommendation

Use managed PostgreSQL 16 (AWS RDS); the production database costs about US\$250–370 a month. We recommend the AWS Malaysia region (about 15% cheaper than Singapore), on db.m7g.large (2 vCPU, 8 GB), Multi-AZ, with 100 GB of storage.

- On-demand: about **US\$315 / month** in Malaysia, US\$369 / month in Singapore.
- Once we know we'll keep it, a 1-year reserved instance (no upfront) brings it to about **US\$249 / month**.
- A test environment adds about US\$69 / month.

Reconciliation needs exact money amounts, no double-counting when the same data is synced again, and matching across tables; PostgreSQL does all three natively. At the estimated volume (about 11 million orders a year), this setup needs no upgrade for 3 years.

Why PostgreSQL

Reconciliation puts 6 hard requirements on the database; PostgreSQL meets all of them natively, with nothing to patch in our own code.

Requirement	Why it matters	How PostgreSQL handles it
Exact money amounts	Floating point leaves 0.01 differences that never reconcile	NUMERIC exact decimals
No double-counting on re-sync	Every 15 minutes we re-pull the last 2 days, so the same order is written again and its status moves from pending to success	Unique key + insert-or-update (ON CONFLICT)
Transactions	A batch of orders and its sync log must succeed or fail together	Full ACID
Cross-table matching	CRM orders match PSP orders by order ID, or by client + amount $\pm 0.5\%$ + 48 hours	SQL JOINs
Keep raw PSP payloads	29 PSPs, all with different fields; audits and later fixes need the original	JSONB, queryable and indexable
Audit and access control	Finance data must show who	Row-level security, pgAudit

Requirement	Why it matters	How PostgreSQL handles it
	changed what	

The schema is already written and the full sync → reconcile → dashboard flow has been tested end to end on PostgreSQL 16.

Data volume estimate

About 11 million orders and 22 GB a year, which is small for PostgreSQL. The estimate uses the business volume in the dashboard prototype: a few hundred to just over a thousand orders per PSP per day.

Item	Estimate
PSPs	29
Orders per day (deposits + withdrawals)	~30,000
Orders per year	~11 million
Storage per year (~2 KB per order incl. raw payload)	~22 GB

Even at 5× this volume the setup below is enough; only storage needs adding earlier, at about US\$25 / month per extra 100 GB.

Cost

The recommended setup in the Malaysia region costs about US\$315 / month on demand, or about US\$249 / month (about US\$3,000 a year) on a 1-year reserved instance. Prices come from AWS's official price list (published 2026-10-06), at 730 hours a month, in US dollars, before tax.

Setup	Use	Malaysia (US/month)	
db.t4g.medium (2 vCPU, 4 GB) Single-AZ, 20 GB	Test environment	69	77
db.t4g.medium (2 vCPU, 4 GB) Multi-AZ, 50 GB	Minimum production	146	162
db.m7g.large (2 vCPU, 8 GB) Multi-AZ, 100 GB	Recommended production	315	369
Same, 1-year reserved (no upfront)	Once we commit long term	249	291

- **Why Multi-AZ:** if one machine fails, a standby takes over automatically, so reconciliation and the screen board keep running. Instance cost is 2× Single-AZ.

- **Why m7g rather than the cheaper t4g:** t4g is burstable; sustained load is billed extra as CPU credits (US\$0.075 per vCPU-hour). Syncing every 15 minutes plus reconciliation is steady load, so m7g keeps cost predictable. On a tight budget, start on t4g and upgrade in one click later.
- **Backups:** free up to the database size; beyond that about US\$0.09 per GB-month.
- **Not included:** the server running the sync jobs and API, and the fixed egress IP registered with PSPs (not priced here).
- **Why not self-host:** installing PostgreSQL on our own cloud server could roughly halve the machine cost, but backups, failover and patching become our job. Not worth it for finance data.
- Alibaba Cloud's Kuala Lumpur region is also an option; its official prices were not checked this time and can be added if needed.

Compared with other options

MySQL is the only workable alternative; the others do not suit a reconciliation database.

Option	Good for	Why not	Cost compared
PostgreSQL (recommended)	Reconciliation database	—	See Cost
MySQL 8	Could also run reconciliation	Weaker at storing raw payloads and conditional aggregation; porting the existing code takes about 1 day	Same AWS price for the same size, no saving
MongoDB	Data without a fixed shape	Reconciliation relies on cross-table matching and exact amounts, which a document store pushes into our code and makes error-prone; PostgreSQL already stores raw payloads	Managed plans are usually no cheaper, plus one more system to run
ClickHouse	Analytics on billions of rows	Poor at updating existing rows, and order status changes; only fits later as an analytics copy	Extra system and staff time we don't need yet
BigQuery / Snowflake	Company-wide data warehouse	Billed per query, so a screen board refreshing every few seconds gets	Grows with usage, monthly bill hard to predict

Option	Good for	Why not	Cost compared
		expensive; higher latency	

Deployment and scaling

Set up these 5 things at launch; add anything else only when the matching situation below appears.

1. **Automatic backups** kept for 35 days, with point-in-time restore on.
2. **Fixed egress IP**: sync jobs reach PSPs from one fixed IP, which we give to PSPs that require whitelisting.
3. **No credentials in the database**: PSP credentials live in AWS Secrets Manager.
4. **Separate accounts**: sync jobs use a write account; the dashboard uses a read-only account.
5. **Run on demand for 1–2 months** to confirm the size, then buy the reserved instance.

When	Add	Approx. cost
Orders table passes 50 million rows and daily queries slow down	Partition the orders table by month (no code change)	Free
Storage nearly full	More storage	~US\$25 / month per 100 GB
Finance wants to write SQL reports directly	A read replica	About the Single-AZ instance price
Screen board traffic grows and the API takes over 1 second	Cache results for 1 minute	Free (done in code first)
Multi-year historical analysis	Daily copy into ClickHouse / BigQuery	Assess at the time

Sources

- [AWS RDS official price list: Singapore, ap-southeast-1](#) (published 2026-10-06)
- [AWS RDS official price list: Malaysia, ap-southeast-5](#) (published 2026-10-06)